ECON-GA 1802, Industrial Organization II
Spring 2026

Description

Instructor: Thi Mai Anh Nguyen, 19 W 4th Street, Room 815 (tmn285@nyu.edu)
Office Hours: By appointment
Lecture: Monday, 12pm-2pm, 19 W 4th Street, Room 624

This course focuses on the empirical analysis of theoretical models of market behavior. In combination with Professors Boyan Jovanovic, Chris Conlon, and Audrey Tiew' graduate IO courses, this course will prepare you to write a dissertation in empirical industrial organization.

We will study how to formulate models, construct empirical specifications that are tractable for estimation, and conduct counterfactual analysis used to answer questions. Throughout, the emphasis will be on understanding how the data and theory can be linked to answer an economic question of interest. The topics covered will focus on two big themes: (i) different modes of transactions (auctions, contracts, relationships, bargaining) and (ii) agency and market frictions (search frictions, asymmetric information, unobservable actions, incomplete contracts).

Although the topics and applications studied focus on issues in industrial organization, the basic methodological framework is likely useful for students in other fields interested in estimating models derived from economic theory.

Assessment:

1 Assignments (50%)

There will be two long problem sets. You will derive testable predictions from models, derive estimators, and conduct empirical and counterfactual exercises that are simplified versions of problems one is likely to encounter when writing a paper in empirical IO.

You must submit your own original code and write-up for each assignment. There is simply no substitute for deriving and coding these estimators yourself. That said, you are welcome and encouraged to discuss the assignments with your classmates. Please list all the students with whom you have discussed the assignment on your problem set submission. The answers are due via the class website. Please typeset your answers and turn in PDF files and your source code.

While I will accept code in any language, you are encouraged to use common programming languages such as R, Julia, MATLAB, or Python (R and Julia are easiest for me). Please write code that is well-documented, follows transparent logic, and uses short, descriptive variable names. Most importantly, ensure that your results can be easily reproduced. I should be able to run your code by calling a main script file that produces any tables, figures, or numbers used in your write-up. If I cannot reproduce your results, you will not receive credit for the assignment.

2 Reading Group (20%)

Format: We will designate time during the semester to discuss some papers as a group. Each reading group discussion will be about 45 minutes and focus on one past or current job market paper. Everyone is expected to participate, but one-three students will lead the discussion.

- Each student: Read the paper and submit a written response by 10am (EST) on the Friday before the group discussion on Monday. These written responses should be short—1-2 pages double-spaced—and should address the following questions:

  (i) What are the key points of the paper?

  (ii) What are the novel conceptual or methodological contributions of the paper?

  (iii) Do you have any concerns about the paper's data, model, empirical approaches, or how the authors interpret their findings? If so, how might you address those concerns?

  (iv) How would you further develop the ideas/methods of the paper? What would be a good setting for your idea?

  If you are not the discussion leaders, write down your questions about the paper in the shared Google document before the discussion.

- Discussion leaders (1-3 students per paper):
  - Read the paper in detail and present the key points of the paper in 15-20 minutes.
  - Create and send out the Google document one week in advance.
  - Prepare to answer questions about the paper and lead the discussion (see points (i)-(iv) above).

Grading:

- 10% from your role as the discussion leader of one paper
- 10% from your participation in the other discussions (including in-class, the Google document, and your written response)

Papers: Before the first class, go through the list of papers below and send me your top five (ranked) choices. Based on the submitted preferences, I will assign papers and discussion leaders.

- Auctions and Search:

  Aspelund, Karl M, and Anna Russo. "Additionality and asymmetric information in environmental markets: Evidence from conservation auctions" (2025).

  Bhattacharya, Vivek. "An empirical model of R&D procurement contests: An analysis of the DOD SBIR program". *Econometrica* 89, no. 5 (2021): 2189–2224.

  Kaye, Aaron. "The personalization paradox: Welfare effects of personalized recommendations in two-sided digital markets" (2024).

  Rosaia, Nicola. "Competing platforms and transport equilibrium". *Econometrica* 93, no. 6 (2025): 2235–2271.

- Asymmetric Information, Moral Hazard, and Contracting:

  Polo, Alberto, Arthur Taburet, and Quynh-Anh Vo. "Screening using a menu of contracts: A structural model for lending markets". *Journal of Financial Economics* 169 (2025): 104056.

  Ryan, Nicholas. "Contract enforcement and productive efficiency: Evidence from the bidding and renegotiation of power contracts in India". *Econometrica* 88, no. 2 (2020): 383–424.

  Xia, Tianli. "Welfare Effects of Resale Price Maintenance: Evidence from the Chinese Pharmaceutical Industry". *Available at SSRN 5118723* (2024).

  Yde, Eric. "Vertical Integration and Regulated Profits in Pharmacy Benefits Markets" (2025, this year's JMP).

  Zahur, Nahim Bin. "Long-term contracts and efficiency in the liquefied natural gas industry". *Available at SSRN 4222408* (2025).

- Platforms:

  Cui, Jingyi. "Bidding for Reputation" (2025, this year's JMP).

  Galdin, Anais, and Jesse Silbert. "Making Talk Cheap: Generative AI and Labor Market Signaling". *arXiv preprint arXiv:2511.08785* (2025, this year's JMP).

  Lee, Kwok Hao, and Leon Musolff. "Two-Sided Markets Shaped by Platform-Guided Search" (2025).

  Sullivan, Michael. "Fee Optimality in a Two-Sided Market" (2025).

Reading Group schedule:

1. SRG 1: 02/23
2. SRG 2: 03/23
3. SRG 3: 04/13
4. SRG 4: 05/04

3 Mini-estimation Project (30%)

Each student is expected to pick (i) a theoretical model or (ii) an empirical model that estimates a theoretically derived model as motivation. Groups of up to three are permissible if the students desire. A theoretical model with only the core elements, maybe with a small twist, should be proposed and an empirical analog should be derived. The next step is to simulate a dataset with parameters from the proposed model. The final step is to write up the estimator and test it on the simulated dataset in a Monte Carlo exercise. Feel encouraged to pick a model that is aligned with your research interests. We hope that this requirement is going to be useful for your future research.

Milestones and respective due dates:

1. 02/23: Select a paper in consultation with me

2. 03/23: Project presentation/model

   Each student will present their project idea in a presentation of at most ten minutes to get feedback from me and the group.

3. 04/20: Simulate data from model

4. 05/04: Estimation and Final Write up

   The final write-up of the project should be about three pages, and no more than five pages. One to two pages for the theoretical and empirical models, one page for the estimator and a page or two describing the results of the exercise. In addition, each student is expected to turn in code that can be run using a main script.

Reading list

The (long) reading list is intended to give you pointers as you explore each topic for your own research. Each lecture will focus on a few papers. Required reading for each lecture will be announced in advance.

A Auctions (Weeks 1-2)

Auctions I

Asker, John. "A study of the internal organization of a bidding cartel". *American Economic Review* 100, no. 3 (2010): 724–762.

Asker, John, and Estelle Cantillon. "Properties of scoring auctions". *The RAND Journal of Economics* 39, no. 1 (2008): 69–85.

Athey, Susan, and Philip A Haile. "Nonparametric approaches to auctions". *Handbook of econometrics* 6 (2007): 3847–3965.

Guerre, Emmanuel, Isabelle Perrigne, and Quang Vuong. "Optimal nonparametric estimation of first-price auctions". *Econometrica* 68, no. 3 (2000): 525–574.

Haile, Philip A, and Elie Tamer. "Inference with an incomplete model of English auctions". *Journal of Political Economy* 111, no. 1 (2003): 1–51.

Laffont, Jean-Jacques, Herve Ossard, and Quang Vuong. "Econometrics of first-price auctions". *Econometrica: Journal of the Econometric Society* (1995): 953–980.

Milgrom, Paul R, and Robert J Weber. "A theory of auctions and competitive bidding". *Econometrica* (1982): 1089–1122.

Auctions II

Aradillas-López, Andrés, Amit Gandhi, and Daniel Quint. "Identification and inference in ascending auctions with correlated private values". *Econometrica* 81, no. 2 (2013): 489–534.

Athey, Susan, and Jonathan Levin. "Information and competition in US forest service timber auctions". *Journal of Political economy* 109, no. 2 (2001): 375–417.

Athey, Susan, Jonathan Levin, and Enrique Seira. "Comparing open and sealed bid auctions: Evidence from timber auctions". *The Quarterly Journal of Economics* 126, no. 1 (2011): 207–257.

Bhattacharya, Vivek. "An empirical model of R&D procurement contests: An analysis of the DOD SBIR program". *Econometrica* 89, no. 5 (2021): 2189–2224.

Campo, Sandra, Isabelle Perrigne, and Quang Vuong. "Asymmetry in first-price auctions with affiliated private values". *Journal of Applied Econometrics* 18, no. 2 (2003): 179–207.

Cassola, Nuno, Ali Hortaçsu, and Jakub Kastl. "The 2007 subprime market crisis through the lens of European Central Bank auctions for short-term funds". *Econometrica* 81, no. 4 (2013): 1309–1345.

Chassang, Sylvain, Kei Kawai, Jun Nakabayashi, and Juan Ortner. "Robust screens for non-competitive bidding in procurement auctions". *Econometrica* 90, no. 1 (2022): 315–346.

Edelman, Benjamin, Michael Ostrovsky, and Michael Schwarz. "Internet advertising and the generalized second-price auction: Selling billions of dollars worth of keywords". *American economic review* 97, no. 1 (2007): 242–259.

Haile, Philip, Han Hong, and Matthew Shum. *Nonparametric tests for common values at first-price sealed-bid auctions*, 2003.

Kastl, Jakub. "Discrete bids and empirical inference in divisible good auctions". *The Review of Economic Studies* 78, no. 3 (2011): 974–1014.

Krasnokutskaya, Elena. "Identification and estimation of auction models with unobserved heterogeneity". *The Review of Economic Studies* 78, no. 1 (2011): 293–327.

Li, Tong, Isabelle Perrigne, and Quang Vuong. "Structural estimation of the affiliated private value auction model". *RAND Journal of Economics* (2002): 171–193.

Li, Tong, and Quang Vuong. "Nonparametric estimation of the measurement error model using multiple indicators". *Journal of Multivariate Analysis* 65, no. 2 (1998): 139–165.

Porter, Robert H, and J Douglas Zona. "Detection of bid rigging in procurement auctions". *Journal of political economy* 101, no. 3 (1993): 518–538.

Roberts, James W. "Unobserved heterogeneity and reserve prices in auctions". *The RAND Journal of Economics* 44, no. 4 (2013): 712–732.

Somaini, Paulo. "Competition and interdependent costs in highway procurement". *Unpublished manuscript.[394, 395]* 2, no. 1 (2011).

B Search and Spatial Search (Weeks 3-5)

Search

Allen, Jason, Robert Clark, and Jean-François Houde. "Search frictions and market power in negotiated-price markets". *Journal of Political Economy* 127, no. 4 (2019): 1550–1598.

— . "The effect of mergers in search markets: Evidence from the Canadian mortgage industry". *American Economic Review* 104, no. 10 (2014): 3365–3396.

Anderson, Simon P, and Regis Renault. "Pricing, product diversity, and search costs: A Bertrand-Chamberlin-Diamond model". *The RAND Journal of Economics* (1999): 719–735.

Choi, Michael, Anovia Yifan Dai, and Kyungmin Kim. "Consumer search and price competition". *Econometrica* 86, no. 4 (2018): 1257–1281.

Coey, Dominic, Bradley J Larsen, and Brennan C Platt. "Discounts and deadlines in consumer search". *American Economic Review* 110, no. 12 (2020): 3748–3785.

Diamond, Peter A. "A model of price adjustment". *Journal of Economic Theory* 3, no. 2 (1971): 156–168.

Ellison, Glenn, and Sara Fisher Ellison. "Search, obfuscation, and price elasticities on the internet". *Econometrica* 77, no. 2 (2009): 427–452.

Gavazza, Alessandro. "The role of trading frictions in real asset markets". *American Economic Review* 101, no. 4 (2011): 1106–1143.

Hong, Han, and Matthew Shum. "Using price distributions to estimate search costs". *The RAND Journal of Economics* 37, no. 2 (2006): 257–275.

Honka, Elisabeth. "Quantifying search and switching costs in the US auto insurance industry". *The RAND Journal of Economics* 45, no. 4 (2014): 847–884.

Hortaçsu, Ali, and Chad Syverson. "Product differentiation, search costs, and competition in the mutual fund industry: A case study of S&P 500 index funds". *The Quarterly Journal of Economics* 119, no. 2 (2004): 403–456.

Salz, Tobias. "Intermediation and competition in search markets: An empirical case study". *Journal of Political Economy* 130, no. 2 (2022): 310–345.

Santos, Babur De los, Ali Hortaçsu, and Matthijs R Wildenbeest. "Testing models of consumer search using data on web browsing and purchasing behavior". *American Economic Review* 102, no. 6 (2012): 2955–2980.

Sorensen, Alan T. "Equilibrium price dispersion in retail markets for prescription drugs". *Journal of Political Economy* 108, no. 4 (2000): 833–850.

Stahl, Dale O. "Oligopolistic pricing with heterogeneous consumer search". *International Journal of Industrial Organization* 14, no. 2 (1996): 243–268.

Spatial Search

Brancaccio, Giulia, Myrto Kalouptsidi, and Theodore Papageorgiou. "Geography, transportation, and endogenous trade costs". *Econometrica* 88, no. 2 (2020): 657–691.

Brancaccio, Giulia, Myrto Kalouptsidi, Theodore Papageorgiou, and Nicola Rosaia. *Search frictions and efficiency in decentralized transportation markets.* Tech. rep. National Bureau of Economic Research, 2020.

Buchholz, Nicholas. "Spatial equilibrium, search frictions and efficient regulation in the taxi industry". *Working paper, Tech. Rep.* (2015).

Castillo, Juan Camilo. "Who benefits from surge pricing?" *Econometrica* 93, no. 5 (2025): 1811–1854.

Frechette, Guillaume R, Alessandro Lizzeri, and Tobias Salz. "Frictions in a competitive, regulated market: Evidence from taxis". *American Economic Review* 109, no. 8 (2019): 2954–2992.

Rosaia, Nicola. "Competing platforms and transport equilibrium". *Econometrica* 93, no. 6 (2025): 2235–2271.

Structural Reading Group 1 (Week 5)

C Contracts (Weeks 6-8)

Selection Markets

Cohen, Alma, and Liran Einav. "Estimating risk preferences from deductible choice". *American economic review* 97, no. 3 (2007): 745–788.

Crawford, Gregory S, Nicola Pavanini, and Fabiano Schivardi. "Asymmetric information and imperfect competition in lending markets". *American Economic Review* 108, no. 7 (2018): 1659–1701.

Cuesta, José Ignacio, and Alberto Sepúlveda. "Price regulation in credit markets: A trade-off between consumer protection and credit access". *Available at SSRN 3282910* (2021).

Decarolis, Francesco, Maria Polyakova, and Stephen P Ryan. "Subsidy design in privately provided social insurance: Lessons from medicare part d". *Journal of Political Economy* 128, no. 5 (2020): 1712–1752.

Einav, Liran, Amy Finkelstein, Stephen P Ryan, Paul Schrimpf, and Mark R Cullen. "Selection on moral hazard in health insurance". *American Economic Review* 103, no. 1 (2013): 178–219.

Einav, Liran, Amy Finkelstein, and Paul Schrimpf. "Optimal mandates and the welfare cost of asymmetric information: Evidence from the UK annuity market". *Econometrica* 78, no. 3 (2010): 1031–1092.

Handel, Ben, Igal Hendel, and Michael D Whinston. "Equilibria in health exchanges: Adverse selection versus reclassification risk". *Econometrica* 83, no. 4 (2015): 1261–1313.

Handel, Benjamin R. "Adverse selection and inertia in health insurance markets: When nudging hurts". *American Economic Review* 103, no. 7 (2013): 2643–2682.

Jin, Yizhou, and Shoshana Vasserman. *Buying data from consumers: The impact of monitoring programs in US auto insurance.* Tech. rep. National Bureau of Economic Research, 2021.

Mahoney, Neale, and E Glen Weyl. "Imperfect competition in selection markets". *Review of Economics and Statistics* 99, no. 4 (2017): 637–651.

Nelson, Scott T. "Private information and price regulation in the us credit card market". *Econometrica* 93, no. 4 (2025): 1371–1410.

Shepard, Mark. "Hospital network competition and adverse selection: evidence from the Massachusetts health insurance exchange". *American Economic Review* 112, no. 2 (2022): 578–615.

Tebaldi, Pietro, Alexander Torgovitsky, and Hanbin Yang. "Nonparametric estimates of demand in the california health insurance exchange". *Econometrica* 91, no. 1 (2023): 107–146.

Vatter, Benjamin. "Quality disclosure and regulation: Scoring design in medicare advantage". *Econometrica* 93, no. 3 (2025): 959–1001.

Contract Design

Bajari, Patrick, and Steven Tadelis. "Incentives versus transaction costs: A theory of procurement contracts". *Rand journal of Economics* (2001): 387–407.

Chade, Hector, Victoria R Marone, Amanda Starc, and Jeroen Swinkels. *Multidimensional screening and menu design in health insurance markets.* Tech. rep. National Bureau of Economic Research, 2022.

Dinerstein, Michael, and Isaac M Opper. *Screening with multitasking: Theory and empirical evidence from teacher tenure reform.* Tech. rep. National Bureau of Economic Research, 2022.

Einav, Liran, Mark Jenkins, and Jonathan Levin. "Contract pricing in consumer credit markets". *Econometrica* 80, no. 4 (2012): 1387–1432.

Holmstrom, Bengt, et al. "Moral hazard and observability". *Bell journal of Economics* 10, no. 1 (1979): 74–91.

Holmstrom, Bengt, and Paul Milgrom. "Multitask principal–agent analyses: Incentive contracts, asset ownership, and job design". *The Journal of Law, Economics, and Organization* 7, no. special_issue (1991): 24–52.

Marone, Victoria R, and Adrienne Sabety. "When should there be vertical choice in health insurance markets?" *American Economic Review* 112, no. 1 (2022): 304–342.

Mussa, Michael, and Sherwin Rosen. "Monopoly and product quality". *Journal of Economic theory* 18, no. 2 (1978): 301–317.

Ryan, Nicholas. "Contract enforcement and productive efficiency: Evidence from the bidding and renegotiation of power contracts in India". *Econometrica* 88, no. 2 (2020): 383–424.

Structural Reading Group 2 (Week 8)

Project Presentation (Week 9)

D Long-Term Contracts and Relationships (Weeks 10-12)

Andrews, Isaiah, and Daniel Barron. "The allocation of future business: Dynamic relational contracts with multiple agents". *American Economic Review* 106, no. 9 (2016): 2742–2759.

Atal, Juan Pablo, Hanming Fang, Martin Karlsson, and Nicolas R Ziebarth. "Long-term health insurance: Theory meets evidence" (2020).

Baker, George P, and Thomas N Hubbard. "Make versus buy in trucking: Asset ownership, job design, and information". *American Economic Review* (2003).

— . "Make versus buy in trucking: Asset ownership, job design, and information". *The Quarterly Journal of Economics* (2004).

Bernheim, B Douglas, and Michael D Whinston. "Multimarket contact and collusive behavior". *The RAND Journal of Economics* (1990): 1–26.

Blouin, Arthur, and Rocco Macchiavello. "Strategic default in the international coffee market". *The Quarterly Journal of Economics* 134, no. 2 (2019): 895–951.

Board, Simon. "Relational contracts and the value of loyalty". *American Economic Review* 101, no. 7 (2011): 3349–3367.

Brugues, Felipe. "Take the goods and run: Contracting frictions and market power in supply chains". *Work. Pap., Brown Univ., Providence, RI* (2020).

Ghili, Soheil, Ben Handel, Igal Hendel, and Michael D Whinston. "Optimal long-term health insurance contracts: characterization, computation, and welfare effects". *Review of Economic Studies* 91, no. 2 (2024): 1085–1121.

Gibbons, Robert. “Four formal(izable) theories of the firm?” *Journal of Economic Behavior & Organization* (2005).

Gibbons, Robert, and John Roberts. *The Handbook of Organizational Economics*. Princeton University Press, 2013.

Harris, Adam, and Thi Mai Anh Nguyen. *Long-term relationships and the spot market: Evidence from us trucking*. Tech. rep. August 2024. Working Paper, 2022.

— . “Long-term relationships in the us truckload freight industry”. *American Economic Journal: Microeconomics* 17, no. 1 (2025): 308–353.

Hubbard, Thomas N. “Contractual form and market thickness in trucking”. *RAND Journal of Economics* (2001).

Joskow, Paul L. “Contract duration and relationship-specific investments: Empirical evidence from coal markets”. *The American Economic Review* (1987): 168–185.

— . “Price adjustment in long-term contracts: the case of coal”. *The Journal of Law and Economics* 31, no. 1 (1988): 47–83.

— . “The performance of long-term contracts: further evidence from coal markets”. *The Rand Journal of Economics* (1990): 251–274.

Kranton, Rachel E. “Reciprocal exchange: A self-sustaining system”. *The American Economic Review* (1996): 830–851.

Levin, Jonathan. “Relational incentive contracts”. *American Economic Review* 93, no. 3 (2003): 835–857.

Macchiavello, Rocco. “Relational contracts and development”. *Annual Review of Economics* 14, no. 1 (2022): 337–362.

Macchiavello, Rocco, and Ameet Morjaria. “Competition and relational contracts in the Rwanda coffee chain”. *The Quarterly Journal of Economics* 136, no. 2 (2021): 1089–1143.

— . “The value of relationships: evidence from a supply shock to Kenyan rose exports”. *American Economic Review* 105, no. 9 (2015): 2911–2945.

MacKay, Alexander. “Contract duration and the costs of market transactions”. *American Economic Journal: Microeconomics* 14, no. 3 (2022): 164–212.

Pavan, Alessandro, Ilya Segal, and Juuso Toikka. “Dynamic mechanism design: A Myersonian approach”. *Econometrica* 82, no. 2 (2014): 601–653.

Sannikov, Yuliy. “A continuous-time version of the principal-agent problem”. *The Review of Economic Studies* 75, no. 3 (2008): 957–984.

Zahur, Nahim Bin. “Long-term contracts and efficiency in the liquefied natural gas industry”. *Available at SSRN 4222408* (2025).

Structural Reading Group 2 (Week 11)

E Vertical Contracts and Bargaining (Weeks 13-14)

Backus, Matthew, Thomas Blake, Brad Larsen, and Steven Tadelis. “Sequential bargaining in the field: Evidence from millions of online bargaining interactions”. *The Quarterly Journal of Economics* 135, no. 3 (2020): 1319–1361.

Berto Villas-Boas, Sofia. “Vertical relationships between manufacturers and retailers: Inference with limited data”. *The Review of Economic Studies* 74, no. 2 (2007): 625–652.

Bolton, Patrick, and Michael D Whinston. “Incomplete contracts, vertical integration, and supply assurance”. *The Review of Economic Studies* 60, no. 1 (1993): 121–148.

Collard-Wexler, Allan, Gautam Gowrisankaran, and Robin S Lee. ““Nash-in-Nash” bargaining: A microfoundation for applied work”. *Journal of Political Economy* 127, no. 1 (2019): 163–195.

Conlon, Christopher T, and Julie Holland Mortimer. “Efficiency and foreclosure effects of vertical rebates: Empirical evidence”. *Journal of Political Economy* 129, no. 12 (2021): 3357–3404.

Crawford, Gregory S, Robin S Lee, Michael D Whinston, and Ali Yurukoglu. “The welfare effects of vertical integration in multichannel television markets”. *Econometrica* 86, no. 3 (2018): 891–954.

Crawford, Gregory S, and Ali Yurukoglu. “The welfare effects of bundling in multichannel television markets”. *American Economic Review* 102, no. 2 (2012): 643–685.

Freyberger, Joachim, and Bradley J Larsen. “How well does bargaining work in consumer markets? A robust bounds approach”. *Econometrica* 93, no. 1 (2025): 161–194.

Ghili, Soheil. “Network formation and bargaining in vertical markets: The case of narrow networks in health insurance”. *Available at SSRN 2857305* (2016).

Grennan, Matthew. “Price discrimination and bargaining: Empirical evidence from medical devices”. *American Economic Review* 103, no. 1 (2013): 145–177.

Ho, Kate, and Robin S Lee. *Contracting over pharmaceutical formularies and rebates*. Tech. rep. National Bureau of Economic Research, 2024.

— . “Equilibrium provider networks: Bargaining and exclusion in health care markets”. *American Economic Review* 109, no. 2 (2019): 473–522.

Ho, Katherine. “Insurer-provider networks in the medical care market”. *American Economic Review* 99, no. 1 (2009): 393–430.

Ho, Katherine, Justin Ho, and Julie Holland Mortimer. “The use of full-line forcing contracts in the video rental industry”. *American Economic Review* 102, no. 2 (2012): 686–719.

Hortaçsu, Ali, and Chad Syverson. “Cementing relationships: Vertical integration, foreclosure, productivity, and prices”. *Journal of political economy* 115, no. 2 (2007): 250–301.

Larsen, Bradley J. “The efficiency of real-world bargaining: Evidence from wholesale used-auto auctions”. *The Review of Economic Studies* 88, no. 2 (2021): 851–882.

Lee, Robin S. “Vertical integration and exclusivity in platform and two-sided markets”. *American Economic Review* 103, no. 7 (2013): 2960–3000.

Ordover, Janusz A, Garth Saloner, and Steven C Salop. “Equilibrium vertical foreclosure”. *The American Economic Review* (1990): 127–142.

Rey, Patrick, and Jean Tirole. “A primer on foreclosure”. *Handbook of industrial organization* 3 (2007): 2145–2220.

Segal, Ilya. “Contracting with externalities”. *The Quarterly Journal of Economics* 114, no. 2 (1999): 337–388.

Segal, Ilya, and Michael D Whinston. “Robust predictions for bilateral contracting with externalities”. *Econometrica* 71, no. 3 (2003): 757–791.

Xia, Tianli. "Welfare Effects of Resale Price Maintenance: Evidence from the Chinese Pharmaceutical Industry". *Available at SSRN 5118723* (2024).

Structural Reading Group 4 (Week 14)